

Scientific Machine Learning

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What to expect...

- What is Scientific Machine Learning?
- How does the course work?
- What do I think is a great talk / report / documentation
- Topics



What is Scientific Machine Learning?

- combination of scientific computing and machine learning, combining mathematical models with data-driven reasoning
- traditional deep learning methodologies have had difficulties with scientific issues like interpretability, enforcing physical constraints, or stiffness of ODEs
- SciML is a field of research with new methods, architectures, and algorithms which overcome these problems while adding the data-driven automatic learning features of modern deep learning
- Many successes: e.g. tools like physics-informed neural networks, universal differential equations, deep backward stochastic differential equation solvers for high dimensional partial differential equations, and neural surrogates

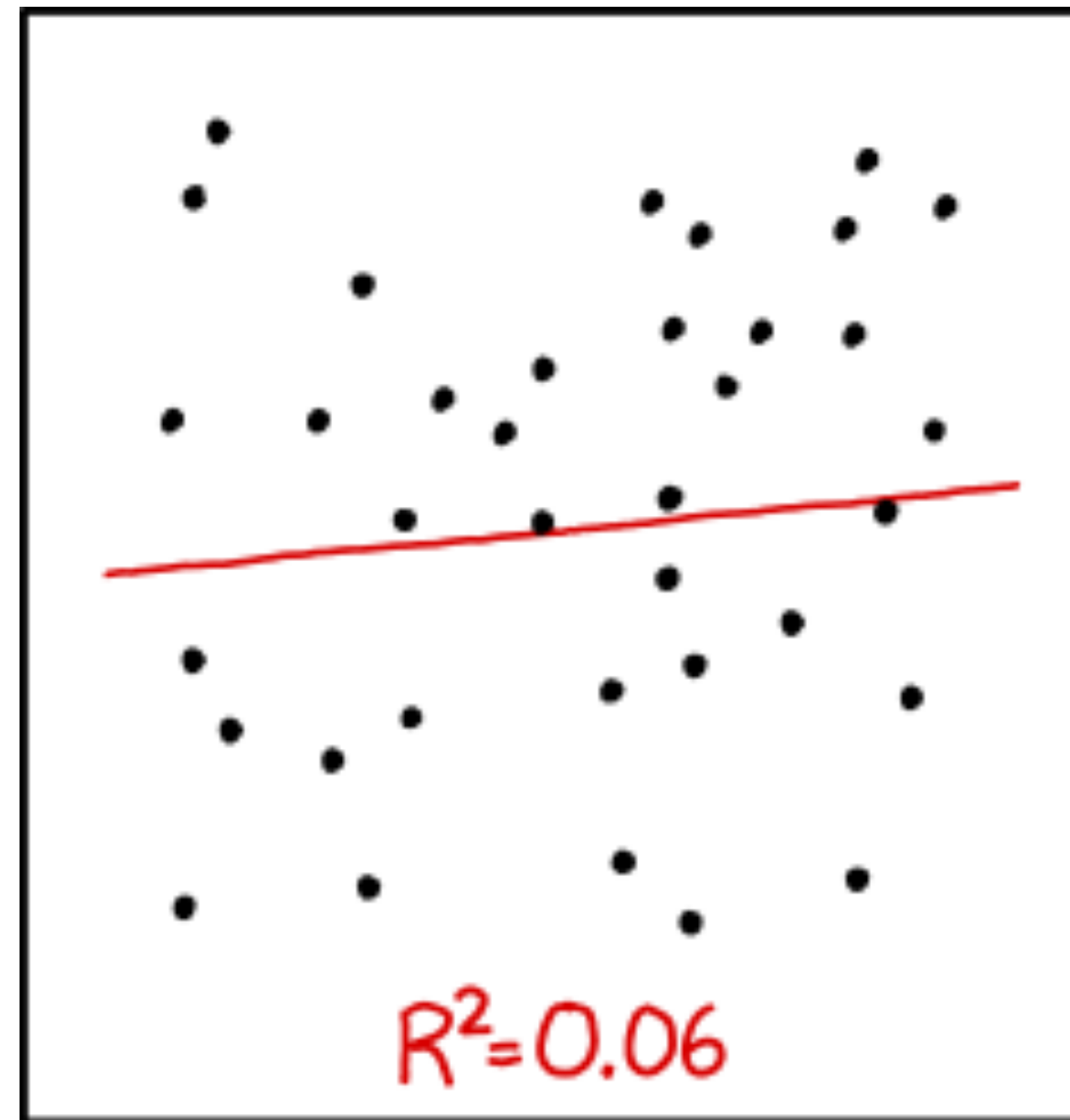
How does the seminar work?

- 13-14 sessions —> 13-14 talks possible
(22./29.4., 6./13./20./27.5., 3./10./17./24.6., 1./8./15./22.7.)
- talks are possible alone or in a team of two, up to you.
- we will have 45 - 60 minutes presentation with additional discussion / questions
I am much in favour of interactive sessions!
- you do the literature search, give a talk and write a report (<5 pages)
- grading is based on the quality of the talk, the written report and participation during the seminar

Seminar structure

From easy to hard / from basic ML background to very ML heavy

- The Basics: important algorithmic details
- From physics/math to ML: Differential equations the classic way and with ML
- generative methods and uncertainty quantification



I DON'T TRUST LINEAR REGRESSIONS WHEN IT'S HARDER TO GUESS THE DIRECTION OF THE CORRELATION FROM THE SCATTER PLOT THAN TO FIND NEW CONSTELLATIONS ON IT.

Topics I: The Basics

- algorithmic details (PyTorch, Tensorflow, Jax and Julia) and differentiable programming: automatic differentiation vs. numerical or symbolic differentiation
- Forward mode and reverse mode automatic differentiation, adjoint derivation for neural ODEs

Topics II: Differential Equations with ML

- Mixing DEs and neural networks for physics informed learning
 - universal differential equations
 - Neural ODEs
- Neural Operators, deep operator learning
- physics informed neural networks

Related Topics...

- equivariant neural networks, hamiltonian neural networks, lagrangian NNs
- symbolic regression

Topics III:

- likelihood-based generative modelling for scientific data
 - Variational Autoencoders
 - normalizing flows
 - diffusion models, flow-matching
- likelihood-free methods
 - simulation-based inference
- model comparison with ML / SBI

Topics III (continued):

- uncertainty quantification
 - bayesian neural networks
 - gaussian processes

What do I think is a good talk/report

- good balance between scientific content, entertainment and level of difficulty
 - give a motivation for the scientific question: Why should your audience care about what you tell us? (So what?)
 - mind your audience!
 - start with the basics, then go into the details, give examples (e.g. code examples or application examples)
 - it is called „giving a talk“ for reasons. Don't use slides as replacement for a report!
 - be critical, conclude and summarize what you (or your audience) have learned
- use this seminar to also learn GitHub, Jupiter and colab